



Exploratory Data Analysis of Fitness & Health Tracking Data

Insights from Activity, Sleep, Calories & Heart Rate Data

Tools: Python, Pandas, NumPy, Matplotlib, Jupyter Notebook

Project Overview & Objectives

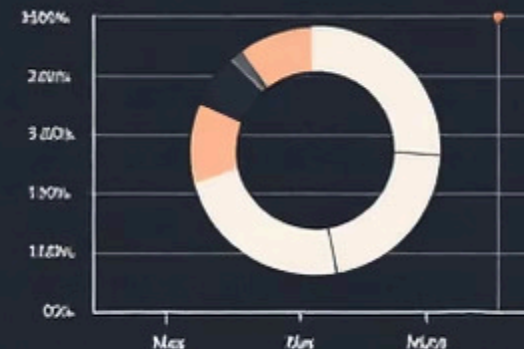
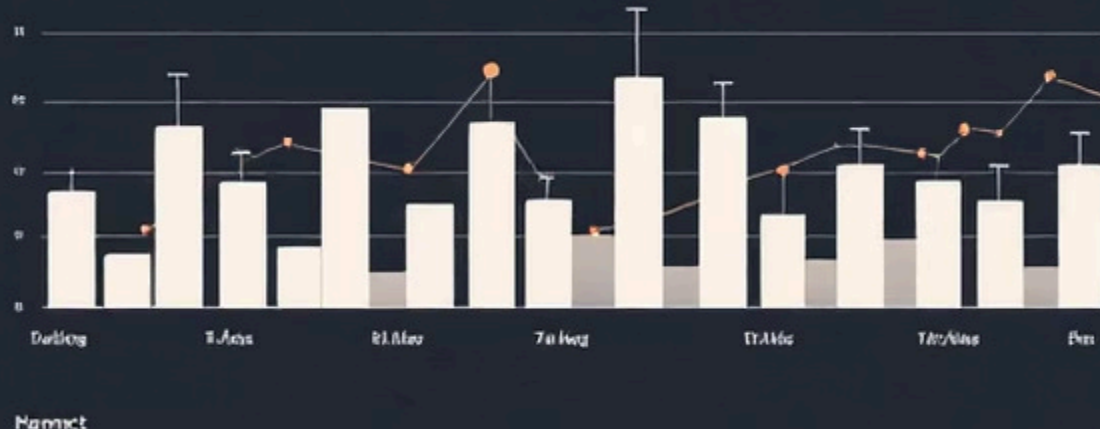


Why This Analysis Matters

Understanding user behavior patterns in fitness tracking helps identify engagement levels, health correlations, and areas for improvement in wellness programs.

Key Questions Explored

- How active are users daily?
- What's the relationship between sleep and activity?
- Are users consistent in tracking?
- How do calories relate to steps?



Dataset Overview

6 Datasets Integrated

Activity, Calories, Steps,
Intensities, Sleep, Heart Rate

33 Users Tracked

Unique participants with wearable
fitness trackers

~30 Days Tracking

Data collected from April to May
2016

Key columns analyzed: Id, Date, Steps, Calories, Active Minutes, Sleep Duration, Heart Rate

Data Cleaning & Preparation Process



Date Standardization

Converted all date fields to consistent format for merging across datasets



Duplicate Removal

Identified and removed duplicate entries based on Id and Date combinations



Missing Data Handling

Sleep data showed highest missing rate—users often forgot to wear trackers overnight



Feature Engineering

Created new features: Day of week, Short sleep indicator (<6 hours)

User Engagement Analysis

Activity Distribution

Users showed varied engagement levels. Some tracked consistently for 30+ days while others had sporadic usage patterns.

Key Finding: Inconsistent users showed lower average activity levels compared to consistent trackers.



33

Total Users

Participants in analysis

~30

Average Days

Per user tracking period

2X

Consistency Gap

Active users vs inconsistent

Activity & Steps Analysis



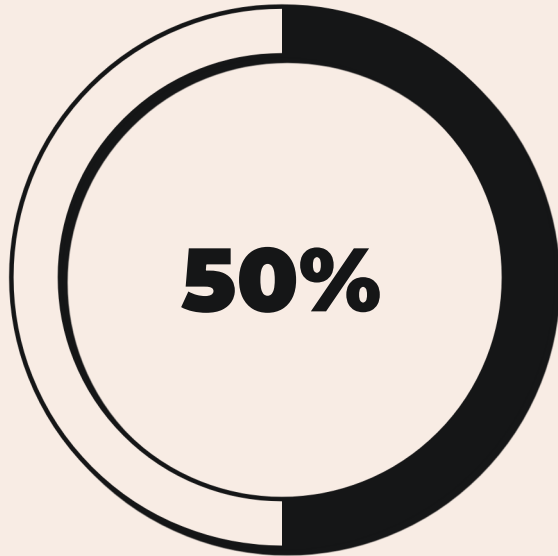
Average Daily Steps

Mean across all users and days



Median Daily Steps

Robust measure of central tendency



Below 7500 Steps

Days falling short of recommended activity

Pro Users

Mean distance > 5 miles

Intermediate

Mean distance 2-5 miles

Beginner

Mean distance < 2 miles



Calories & Sleep Analysis

Calories Burned

Average daily: ~2300 calories

Highest day: April 23, 2016

Some days showed high calories with low steps—likely intense workouts

Sleep Patterns

Average duration: ~7 hours

Missing data: ~56% of days

Users with <6 hours sleep showed 15% lower activity levels

Heart Rate & Weekly Patterns

Heart Rate Analysis

Average resting heart rate: ~72 BPM

Higher heart rates correlated with increased activity minutes. Daily aggregation showed consistent morning resting rates.

Weekday vs Weekend

Weekend activity: ~8,200 steps

Weekday activity: ~7,400 steps

Weekends showed 11% higher step counts—users more active during leisure time

Correlation Insights & Key Findings



Steps & Calories

Positive correlation observed: more steps led to higher calorie burn. Average ~0.3 calories per step.



Sleep & Activity

Clear relationship: Users sleeping ≥ 6 hours showed significantly higher activity levels vs those with short sleep.



BMI & Steps

Negative correlation: Higher BMI users showed lower total step counts over tracking period.

Business Implications & Future Scope

Current Challenges

- 56% sleep data missing
- Inconsistent tracking patterns
- Limited demographic information

Business Value

Companies can use these insights to design personalized wellness programs, identify engagement drop-offs, and create targeted health interventions based on user segments.

Future Enhancements

1. Machine learning predictions for activity levels
2. Personalized activity recommendation systems
3. Long-term health impact tracking
4. Integration with dietary data

📌 **Conclusion:** EDA revealed clear patterns in user behavior, sleep-activity relationships, and engagement levels. Analysis provides actionable insights for fitness programs and health interventions.